

Building a Spotify Recommendation Engine

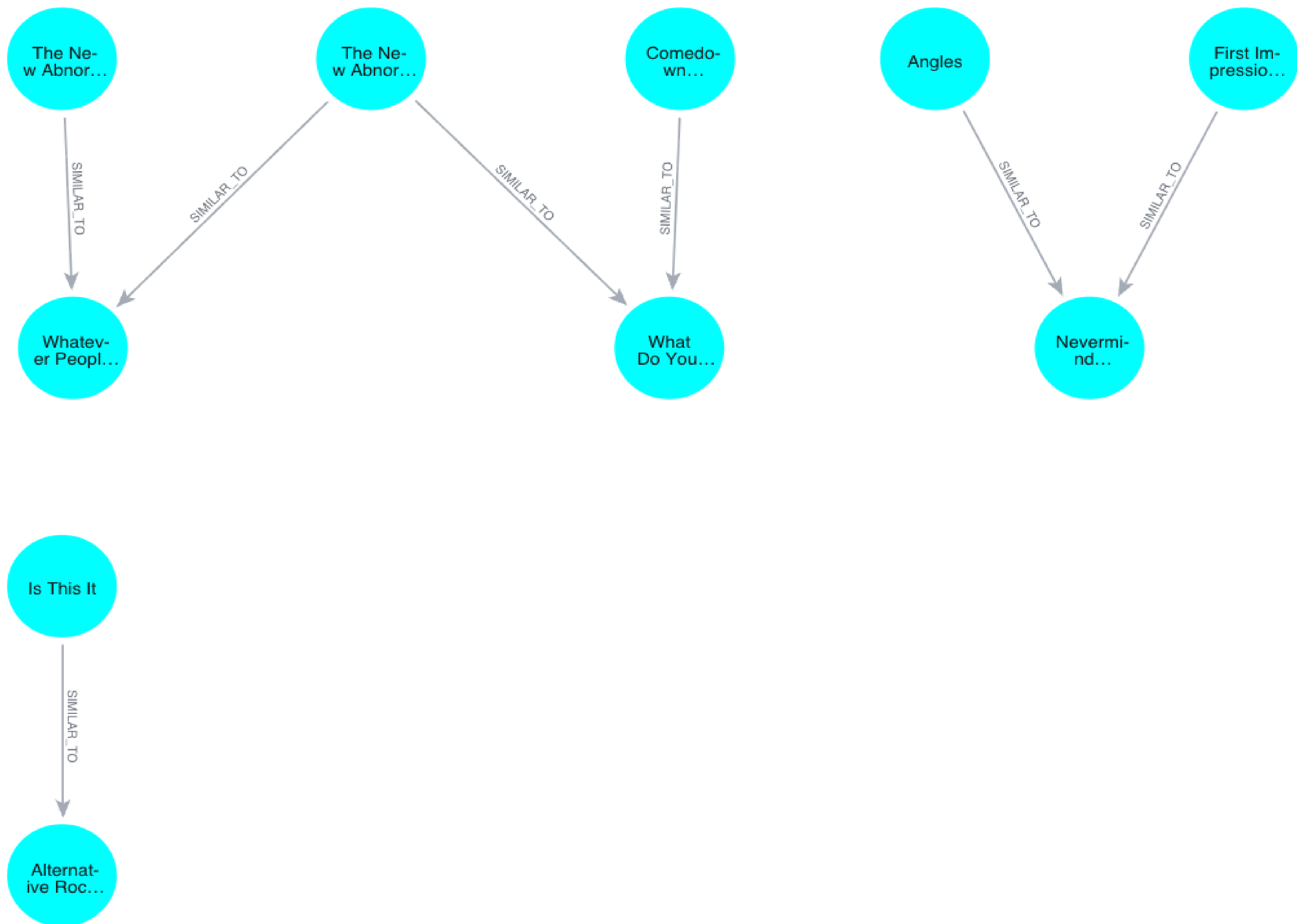
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Approach

We developed a recommendation system where each song is a node and the similarity between songs are the edges. We defined similarity as the Euclidean distance over various audio features, including danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo. Using a k-nearest neighbors' approach with k=10, each song is connected to its closest neighbors. Our final recommendation was generated by starting from songs by *The Strokes* and *Regina Spektor* and identifying nearby songs in that graph that are not by those artists. Results are ranked based on average distance, with a genre filter to ensure stylistic consistency.

Recommendation Graph Model



Nodes & Edges

1046 nodes & 10460 edges

Song Similarity Metric

Euclidean distance over standardized audio features:

danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo

Recommended Songs

1. Brazil — Declan McKenna (*What Do You Think About the Car?*)
2. Mardy Bum — Arctic Monkeys (*Whatever People Say I Am, That's What I'm Not*)
3. Come As You Are — Nirvana (*Nevermind*)
4. Wake Me Up When September Ends — Green Day (*American Idiot*)
5. Million Little Problems — The Walters (*Try Again*)